Research Group for Machine Learning in Cancer Genetics and Precision Medicine

Estimate Mutational Signature Exposure from Sparse Clinical Sequencing Data

Highlighting the merits of inferring Mutational Signatures from short panel data vs whole genome coverage

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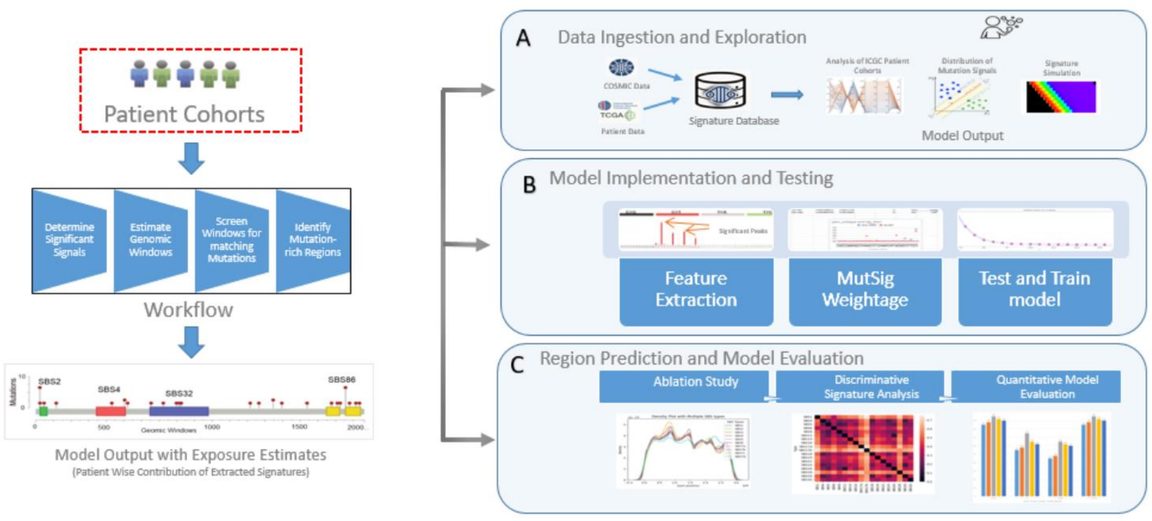
Introduction

The study of mutational signatures based on context-specific mutations has become an important aspect of cancer genome research. *Mutational Signatures* have become an important tool for understanding factors contributing to cancer development and hold promise for developing new diagnosis, treatment, and prevention strategies. Current approaches rely on a large number of mutations to accurately estimate mutational signature exposure. Making this analysis possible when only sparse mutation data are available, such as data generated from panel sequencing or samples with low mutational burden, requires novel developments in the current methodologies for estimating mutational signature exposures. In our work, we propose a novel approach of assessing the presence of mutational signatures from sparse sequencing data.

Research Objectives

- ✓ Determining the ***significant signals*** for each signatures in the COSMIC mutational signature catalog
- ✓ Determine the underlying ***correlations*** among different mutational signatures and making them more discriminative
- ✓ Highlight the ***mutational burden*** for different cancer types in the genome
- ✓ Identify ***informative regions*** of the genome that have a high number of context-specific mutations
- ✓ Determine the effectiveness of our ***"Sparse-Window" based modelling*** approach in extracting mutations without significant information loss

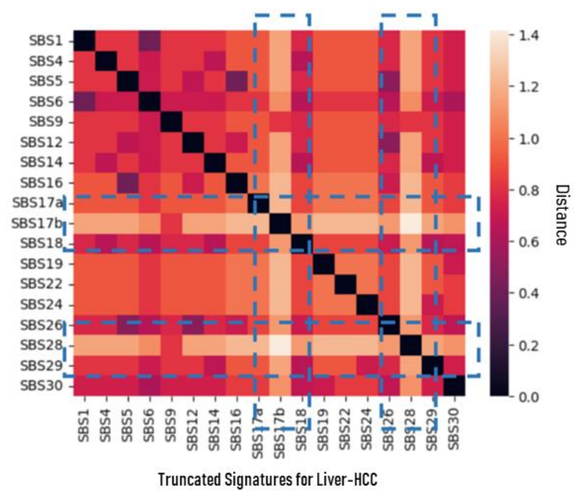
Proposed Model

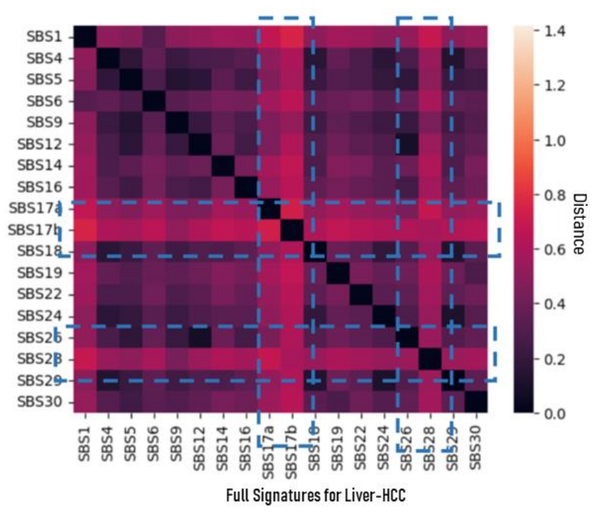


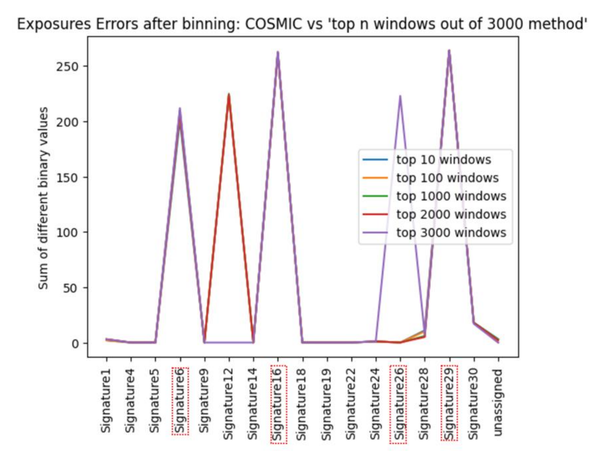
Workflow

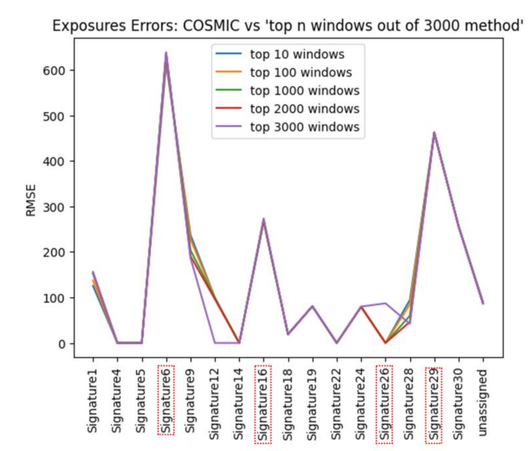
- ➢ Integrate and explore cancer genome data obtained from various patient cohorts
- ➢ Determine the statistically significant signals leading to the elimination of redundant and uninformative signals
- ➢ Build model by introducing the concept of "informative windows"
- ➢ Determine specific informative regions within the genomic sequence that have the highest probability of occurrence for cancer mutations with the help of our trained model
- ➢ Evaluate our model with the ground truth from COSMIC dataset by comparing the exposure estimations.

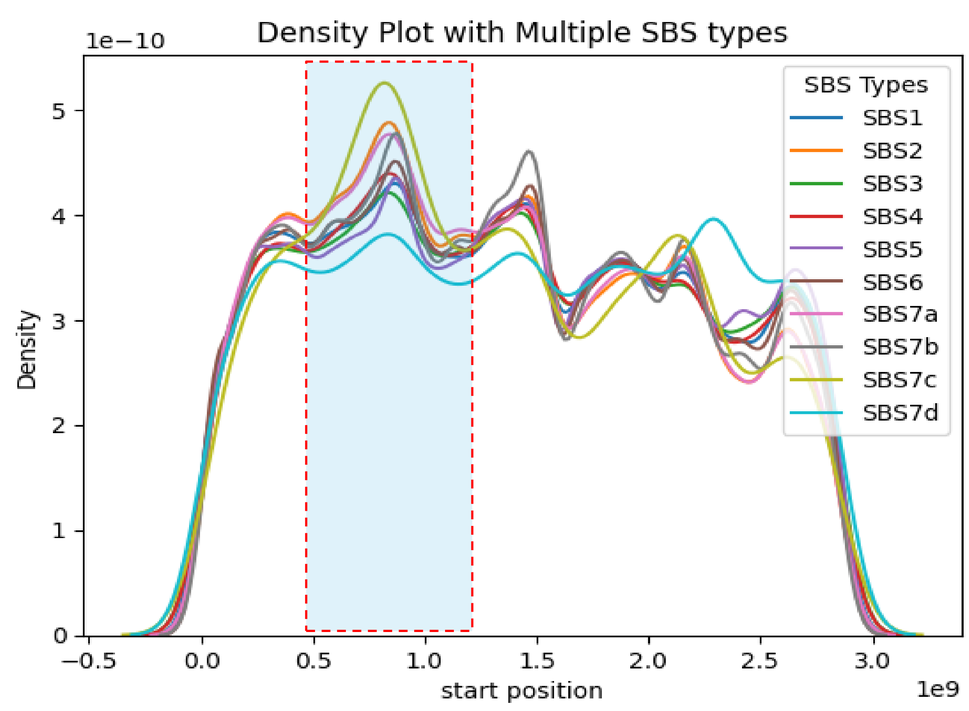
Results











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ISMB 2024, General Computational COSI Track
Montreal, July 12-16 2024